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THE MECHANISM BEHIND MOTHER'S NEGATIVE PARENTING STYLE ON COLLEGE STUDENTS' EMOTIONAL HEALTH

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This study aims to investigate the mechanisms underlying parenting styles on college student's mental health. 935 college students aged 19–21 years were surveyed using the short-form of Egna Minnen Betraffande Uppfostran (S-EMBU), the Depression Anxiety Stress Scales (DASS-21), and the Acceptance and Action Questionnaire-2nd Edition (AAQ-II), to examine the effects of parenting styles on negative emotions in college students and the mediating effect of experiential avoidance. The results showed that: (1) Parenting style can affect experiential avoidance and negative emotions; (2) Experiential avoidance has a partial mediating effect between parenting style and negative emotions; (3) Father's negative parenting style and experiential avoidance have a chain mediating effect on mother's negative parenting style and negative emotions. These findings suggest that parenting styles have a direct effect on the negative emotions of college students and an indirect effect through experiential avoidance.

Key words: parenting style, experiential avoidance, emotional health, mechanism, college students

Introduction

With rapid economic and social development, the mental health of adolescents has been a wide concern. Indicators of negative psychological health, such as depression, anxiety, and stress, are some of the main ways to assess and differentiate mental health status. Studies have shown that common emotional disorders related to depression, anxiety, and stress are widespread among college students, recent research showed that both general mental health and the odds of being diagnosable with a psychiatric disorder have increased among individuals aged 17–24 years old (Tabor et al., 2021). 84% of students with anxiety or depression perceived a need for mental health services (Auty et al., 2022). These emotional disorders have a negative impact on their personality development, academic achievement, and interpersonal relationships. Therefore, it is necessary to explore the mechanisms impacting the mental health of college students and its intervention path.

There are many factors that affect college students' mental health. One of the most

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important factors is significantly influenced by parental behaviors in early life or perceived parenting styles (Sari & Ozkan, 2016). Parenting style refers to the educational concepts, attitudes towards children, and specific parenting behaviors that parents uphold when raising children. Such parenting behaviors are relatively stable and will not change with the environment. Darling and Steinberg (1993) proposed a situational model of parenting style based on psychodynamics, learning theory, and belief system model. Perris et al. (1980) believed that parenting style included four dimensions: rejection, emotional warmth, overprotection, and preference, and the connotation of parenting style varied with different cultural backgrounds. For example, when China implemented the one-child family planning policy in the early years, there was no dimension of preference for the one-child family structure. Most of the college students who participated in the study were children with no siblings and were unable to answer this question. Therefore, the Chinese version of the parenting style questionnaire only contains three dimensions: rejection, emotional warmth, and overprotection (Jiang et al., 2010).

Different parenting styles lead to varied developmental outcomes for adolescents. Positive parenting styles include emotional warmth, support, and encouragement shown by parents in the process of parenting (Li et al., 2019), while negative parenting styles refer to rejection, despotism, punishment, and over-protection shown by parents in the process of parenting (Baardstu et al., 2017). Recent studies have revealed that parenting style in family education affects the development of children's psychological behavior. It has significant effects on psychological traits and behaviors such as happiness (Asici & Sari, 2021), narcissism (Kılıçkaya et al., 2023), and trait emotional intelligence (Argyriou et al., 2016). A harsh/coercive parenting style was strongly associated with emotionally-dysregulated children (Coplan et al., 2009). The influence of authoritarian parenting style on children's wellbeing was sustained during emerging adulthood (Parra et al., 2019). A study found that perceived parental psychological pressure correlated positively with depersonalisation and trait anxiety among adolescents. Perceived parental warmth was positively associated with active coping and negatively correlated with trait anxiety in adolescents (Wolfradt et al., 2003). Another study pointed out the relationship between parenting styles and negative emotion differentiation (Starr et al., 2020). Parents' emotional support can enhance adolescents' sense of self-worth and security, and improve their mental health (Coplan et al., 2002; Steinberg, 2001). Parenting styles have an important impact on children's mental health (Rothrauff et al., 2009). Depression, anxiety, and pressure are three common negative emotions, which are important indicators of mental health (Lovibond & Lovibond, 1996). To further explore the effect of negative parenting style on negative emotions, Hypothesis 1 is proposed: Parenting style directly affects negative emotions.

Acceptance and Commitment Therapy (ACT), developed in the early 1990s, is known as the "third wave" of cognitive behavioral therapy (Hayes, 2004). Its core concept is psychological flexibility, which refers to the individual's conscious experience of the present moment. It is a psychological trait that constantly adjusts one's perspective according to one's value orientation, to adapt to the changing internal and external environment, and focus on valuable actions to achieve certain life goals and self-worth

(Hayes et al., 2006). Based on the basic concept of psychological flexibility and the analysis of a large number of clinical cases, Hayes et al. (2006) proposed the connotation structure of psychological flexibility, which includes six aspects: Acceptance; Defusion; Self as Context; Contact with the Present Moment; Values; Committed Action. The concept corresponding to psychological flexibility is psychological inflexibility, which refers to psychological characteristics that prevent individuals from adapting to the internal and external environment and lead to unnecessary suffering. The psychological inflexibility model also includes six aspects: Experiential Avoidance; Cognitive Fusion; Attachment to the Conceptualized Self; Dominance of the Conceptualized Past and Feared Future; Weak Self-Knowledge, Lack of Values Clarity; Dominance of Pliance and Avoidant Tracking, Inaction, Impulsivity, or Avoidant Persistence. Hayes believes that psychological inflexibility is the root cause of individual pain and functional disorders, and experiential avoidance is the main variable of the psychological inflexibility model (Coplan et al., 2002; Hayes, 2016; Monestès et al., 2018). As the main variable of the psychological inflexibility model, "experiential avoidance" corresponds to "acceptance" in the psychological flexibility model. Experiential avoidance occurs when a person is unwilling to stay in touch with specific personal experiences (e.g., bodily sensations, emotions, thoughts, memories, behaviors) and seeks to change them. For example, if a person experiences a negative event in early childhood, the person will deliberately avoid relevant negative information as an adult to reduce the damage of the negative event. Experiential avoidance is associated with the growth experiences of individuals and negatively affects their mental health (Hayes et al., 2004). Studies have shown that the parenting style in individual memory is closely related to life development (Rothrauff et al., 2009), and hence suggest that parenting style will affect experiential avoidance.

Parenting styles influence psychological flexibility, personality development, mental health, and academic level (Fang et al., 2003; Wang & Fu, 2005; H. Zhang, Tian, et al., 2021). Furthermore, adolescent psychological flexibility may affect negative emotions and behavioral styles (Alonso-Stuyck, 2019). Authoritarian parenting styles such as disciplinary practices may cause social avoidance and suppression (McKinney et al., 2011). However, whether parenting style affects mental health through individual experiential avoidance remains unclear. The developmental contextualism theory points out that individual psychological development is not only affected by endogenous factors, but also by the environment or experience, and more importantly, is the result of the interaction between individuals and situations (Lerner & Busch-Rossnagel, 1981). The influence mechanism of four factors, such as physical environment, social members, developing individuals, and situational variables changing over time, on individual development (Lerner et al., 2023; Zhang & Chen, 2009) put forward the view that the development process promotes itself, that is, the individual is the architect of its development (Lerner & Busch-Rossnagel, 1981). This provides a great possibility for the initiative of individual psychological development and also provides a theoretical basis for whether empirical avoidance can play a mediating role between parenting styles and negative emotions. Therefore, Hypothesis 2 is proposed in this study: Empirical avoidance plays a mediating role in the influence of parenting styles on negative emotions.

The parenting styles adopted by fathers and mothers are different due to their role orientations (Paquette, 2004; Shan & Hawkins, 2014). Parenting style may vary according to the gender of the parent. A study found that maternal rejection prevented adolescents from developing emotional self-efficacy; however, paternal rejection did not have such an impact (Niditch & Varela, 2012). A study conducted with Italian adolescents noted that mothers' autonomy-supporting attitudes positively affected the development of trait emotional intelligence, while fathers' attitudes had no significant effect (Gugliandolo et al., 2019). The Children's Work Department of Chinese National Woman Association Union (2015) held a conference to release the main results and core data of their survey. The survey report revealed that mothers play a main role in the division of labor in family education. In addition, mothers' parenting styles may have an impact on fathers' parenting styles. Most studies have comprehensively explored the relationship between parenting styles and other variables, while the relationship between mothers' and fathers' parenting styles has been rarely explored. According to the family system theory, the family is an interconnected system with complex interaction patterns and relationship networks among family members. Each member in the family system does not exist in isolation, and their behaviors and emotions will affect other family members, forming an interdependent dynamic relationship (Bowen, 1993). Hence, we posit that the different parenting styles of fathers and mothers can affect each other, as proposed in Hypothesis 3: Father's parenting style and experiential avoidance have a chain mediation effect on the influence of maternal parenting style on negative emotions.

Methods

Participants and Procedures

Participants were recruited from a university in China. No prior sample size calculations were carried out, and convenient sampling was used to select students. The questionnaires were distributed online to students through college counselors. Invalid questionnaires with incomplete and regular answers were deleted, resulting in 935 valid questionnaires, with an effective recovery rate of 96.2%. The average age of the selected students was 19.7 years (SD = 1.2 years); 562 (60.1%) were male, 373 (39.9%) were female; 468 (50.1%) students had left-behind experience, and 467 (49.9%) students had no left-behind experience.

All procedures performed in the current study involving human participants were in accordance with the ethical standards of the Ethical Committee of Anhui Normal University and with the 1964 Helsinki Declaration and its later amendments or comparable ethics. Informed consent was obtained from all participants included in the study.

Measures

The short-form of Egna Minnen Betraffande Uppfostran. The Egna Minnen Betraffande Uppfostran (EMBU) was first developed by Swiss scholars (Perris et al., 1980) to examine parenting style by asking participants to recall the way their parents treated them when they were growing up. In this study, the revised short-form EMBU (S-EMBU-C) was used. The 42 items in S-EMBU-C were divided into father's and mother's parts, with 21 items in each part. The items were consistent and had 3 dimensions: rejection (1, 4, 7, 12, 14, 19), emotional warmth (2, 6, 9, 11, 13, 17, 21), and overprotection (3, 5, 8, 10, 15, 16, 18, 20). Referring to existing research paradigms (Cai et al., 2022; Li, 2018), positive parenting styles in this study were rated by scores on the emotional warmth dimension, and negative parenting styles were rated by scores on the rejection and overprotection dimensions. The questionnaire was based on a 4-point scale, from

1 (never) to 4 (always), with the 15th item scored in reverse. The scores of the items on the different dimensions are added together, with higher scores indicating that the individual is at a higher level on that dimension and vice versa. In this study, the Cronbach's alphas were .83 for rejection, .90 for emotional warmth, and .71 for overprotection for fathers; .82 for rejection, .90 for emotional warmth, and .72 for overprotection for mothers.

Depression Anxiety Stress Scales. Depression Anxiety Stress Scales (DASS-21) was complied by Lovibond and Lovibond (1996), and a simplified Chinese version of the DASS-21 was introduced by Gong et al. (2010) in the study of Chinese college students. The 21-item scale contained 3 subscales of depression, anxiety, and stress. Furthermore, it was scored on a 4-point scale, ranging from 0 (not at all) to 3 (entirely). The scores of all items were summed, with higher scores indicating more negative emotional experiences. In this study, the Cronbach's alphas for the 3 subscales were .87 for depression, .83 for anxiety, and .85 for stress, with a total Cronbach's alpha of .94.

Acceptance and Action Questionnaire-2nd Edition. The Acceptance and Action Questionnaire-2nd Edition (AAQ-II) was developed by Bond et al. (2011) and translated by Cao et al. (2013) in the study of Chinese college students. The questionnaire had 7 items and was scored on a 7-point scale, from 1 (never) to 7 (always). The items were as follows: (1) My painful experiences and memories make it difficult for me to live a life that I would value; (2) I'm afraid of my feelings; (3) I worry about not being able to control my worries and feelings; (4) My painful memories prevent me from having a fulfilling life; (5) Emotions cause problems in my life; (6) It seems like most people are handling their lives better than I am; (7) Worries get in the way of my success. The scores of all items were summed, with higher scores indicating higher levels of experiential avoidance. In this study, Cronbach's alpha for AAQ-II was .91.

Data Analysis

Data processing and statistical analysis were performed using SPSS 24.0, including Harman's single factor test, reliability analysis, Pearson's correlation coefficient, and regression analysis. In addition, the mediating effect was tested using the Process model, and the significance of the mediating effect was tested using the Nonparametric Bootstrap method.

Results

Common Method Bias Test

Harman's single-factor test was used in this study to conduct exploratory factor analysis (EFA; unrotated) for all original items. The test results showed 13 factors with eigenvalues greater than 1. The maximum eigenvalue was 17.65, and the maximum explained variance was 25.22%. According to Podsakoff and Organ (1986), it is generally believed that the variance explained by a single factor cannot exceed 40%. Therefore, there is no serious common method bias in this study.

Descriptive Statistics and Correlations

Analysis of differences in demographic characteristics. There were significant differences between male and female students in the scores of father's positive parenting style, mother's positive parenting style, father's negative parenting style, mother's negative parenting style, and experiential avoidance. Furthermore, there was no significant difference in the scores of negative emotions. Left-behind experiences were significantly different for father's positive parenting style, mother's positive parenting style, negative emotions, and experiential avoidance scores. In contrast, the left-behind experience was not significant for father's negative parenting style and mother's negative parenting style

Table 1. Analysis of Differences Between Demographic Characteristics (N = 935)

Variables	Male students	Female students	t	Left-behind experience	No left-behind experience	t
	$M \pm SD$	$M \pm SD$		$M \pm SD$	$M \pm SD$	
Father's positive parenting style	18.47 ± 4.72	19.58 ± 4.84	–3.50***	18.06 ± 4.58	19.76 ± 4.86	–5.50***
Mother's positive parenting style	18.94 ± 4.73	20.38 ± 4.56	–4.62***	18.73 ± 4.59	20.31 ± 4.71	–5.19***
Father's negative parenting style	25.11 ± 5.26	23.82 ± 5.13	3.70***	24.86 ± 5.14	24.33 ± 5.34	1.56
Mother's negative parenting style	25.64 ± 5.28	24.69 ± 5.44	2.67**	25.58 ± 5.27	24.94 ± 5.44	1.85
Negative emotions	34.85 ± 10.28	33.63 ± 10.28	1.80	35.43 ± 10.40	33.29 ± 9.83	3.24**
Experiential avoidance	33.63 ± 9.96	20.87 ± 8.20	–2.92**	21.01 ± 7.81	18.90 ± 7.78	4.14***

Note. Left-behind experience refers to the following experiences during the period from birth to 16 years of age: both parents or one parent went out to work while some participants stayed in their hometown without living with them, and the duration lasted for more than half a year.
**p < .01, ***p < .001.

(Table 1).

Correlation Analysis

The results of Pearson's correlation analyses show that: (1) Father's positive parenting style is positively correlated with mother's positive parenting style, negatively correlated with mother's negative parenting style, negative emotions, and experiential avoidance; (2) Mother's positive parenting style is negatively correlated with father's negative parenting style, negative emotions, and experiential avoidance; (3) Father's negative parenting style is positively correlated with mother's negative parenting style, negative emotions, and experiential avoidance; (4) Mother's negative parenting style is positively correlated with negative emotions, experiential avoidance; (5) Negative emotions are positively correlated with experiential avoidance. All correlations were statistically significant, as shown in Table 2.

Tests for Mediating Effects

Mediating effects of experiential avoidance in parenting style on negative emotions. A linear regression was conducted where positive and negative parenting styles of fathers and mothers were used as predictive variables, experiential avoidance and negative emotions were used as dependent variables, controlling for gender and left-behind experiences. The results revealed that positive parenting styles negatively affected experiential avoidance and negative emotions, and negative parenting styles positively

Table 2. Descriptive Statistics and Correlations for Each Variable (N = 935)

Variables	$M \pm SD$	1	2	3	4	5	6
1 Father's positive parenting style	18.91 ± 4.79	1					
2 Mother's positive parenting style	19.52 ± 4.72	.88***	1				
3 Father's negative parenting style	24.60 ± 5.24	–.22***	–.26***	1			
4 Mother's negative parenting style	25.26 ± 5.36	–.24***	–.25***	.85***	1		
5 Negative emotions	34.36 ± 10.17	–.34***	–.37***	.42***	.38***	1	
6 Experiential avoidance	19.95 ± 7.87	–.18***	–.17***	.28***	.30***	.59***	1

***p < .001.

Table 3. Linear Regression Results for Parenting Style, Experiential Avoidance, and Negative Emotions (N = 935)

Parenting style	Experiential avoidance			Negative emotions		
	β	R^2	t	β	R^2	t
Father's positive parenting style	–.30	.03	–5.61***	–.71	.11	–10.93***
Mother's positive parenting style	–.29	.03	–5.36***	–.79	.14	–12.06***
Father's negative parenting style	.43	.08	9.05***	.82	.18	14.27***
Mother's negative parenting style	.44	.09	9.47***	.72	.15	12.63***

***p < .001.

affected experiential avoidance and negative emotions (Table 3).

The SPSS plug-in, Process (model 4; Hayes, 2013), was used with father's and mother's positive and negative parenting styles as predictive variables, experiential avoidance as a mediating variable, and negative emotions as the dependent variable. Based on the mediating effect test procedure proposed by Shrout and Bolger (2002), the iterations was set at 5000, and the significance of the mediating effect was tested by the Nonparametric Bootstrap method at 95% CI. The proportion of the indirect effect to the total effect is calculated by dividing the indirect effect value by the total effect value. The results show that:

(1) The indirect effect of the pathway with father's positive parenting style as the predictor, negative emotions as the dependent variable, and experiential avoidance as the mediating variable is –.21, 95% CI [–.29, –.13], and the direct effect is –.48, p < .001. This result suggests that experiential avoidance partially mediates the effect of father's positive parenting style on negative emotions, with indirect effect accounting for 30% of the total effect.

(2) The indirect effect of the pathway with mother's positive parenting style as the

predictor, negative emotions as the dependent variable, and experiential avoidance as the mediating variable is –.21, 95% CI [–.30, –.13], and the direct effect is –.57, $p < .001$. This result indicates that experiential avoidance partially mediates the effect of mother's positive parenting style on negative emotions, with indirect effect accounting for 27% of the total effect.

(3) The indirect effect of the pathway with father's negative parenting style as the predictor, negative emotions as the dependent variable, and experiential avoidance as the mediating variable is .30, 95% CI [.23, .37], and the direct effect is .51, $p < .001$. This result indicates that experiential avoidance partially mediates the effect of negative emotions in father's negative parenting styles, with indirect effects accounting for 37% of the total effect.

(4) The indirect effect of the pathway with mother's negative parenting style as the predictor, negative emotions as the dependent variable, and experiential avoidance as the mediating variable is .31, 95% CI [.23, .38], and the direct effect is .41, $p < .001$. This result suggests that experiential avoidance partially mediates the effect of mother's negative parenting style on negative emotions, with the indirect effect accounting for 44% of the total effect.

Chain mediation effect of father's parenting style and experiential avoidance on the effect of mother's parenting style on negative emotions. The SPSS plug-in Process (model 6) was used, with mother's negative parenting style as the predictor, negative emotions as the dependent variable, father's negative parenting style and experiential avoidance as chain medication variables, and gender and experience of remaining behind as control variables. The number of iterations was set at 5000, and the chain mediation effect was tested at 95% CI using Bootstrap sampling. The results reveal that the indirect effect of the pathway with father's negative parenting style as a mediating variable is .46, 95% CI [.30, .62], and the indirect effect of the pathway with experiential avoidance as a mediating variable is .18, 95% CI [.07, .30]. In addition, the indirect effect of the

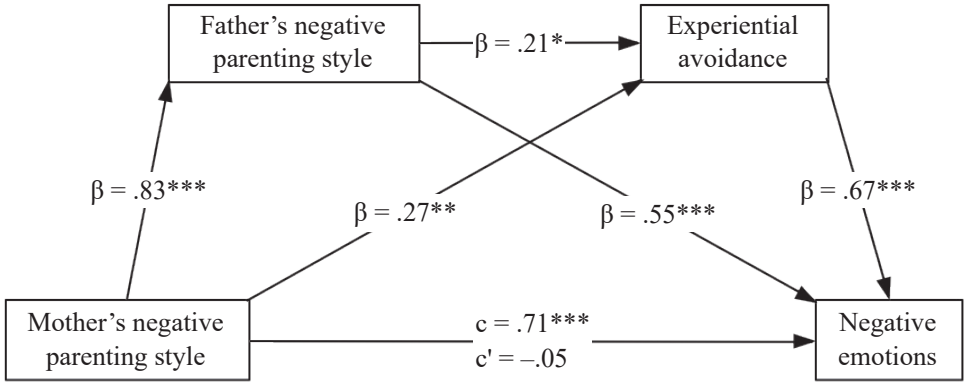


Fig. 1. The Chain Mediation Effect of the Father's Negative Parenting Style and Experiential Avoidance in the Effect of the Mother's Negative Parenting Style on Negative Emotions

Note. c = total effect; c' = direct effect.

*$p < .05$, **$p < .01$, ***$p < .001$.

pathway with father's negative parenting style and experiential avoidance as mediating variables is .12, 95% CI [.03, .22]. The total of all indirect effects is .76, 95% CI [.58, .93]. These results demonstrate the chain mediation effect of the father's negative parenting style and experiential avoidance in the effect of the mother's negative parenting style on negative emotions (coefficient results of pathways are shown in Fig. 1).

Discussion

The results show that parenting style was associated with negative emotional experiences and mental health among college students, supporting Hypothesis 1. Specifically, positive parenting styles (emotional warmth) can reduce depression-anxiety-stress levels in college students, and negative parenting styles (rejection, overprotection) can increase depression-anxiety-stress levels. This is consistent with relevant research in the Chinese cultural context (Li et al., 2003; Wen et al., 2021; J. R. Zhang, Meng, et al., 2021), and is a confirmation of the reproducibility of previous studies. The mental health of college students is jointly influenced by social, family, and individual factors. Among various influencing factors, family education has a profound impact on students. As teachers of their children, parents play a crucial role in their children's growth and development, especially through their parenting style. In the historical context of the one-child policy, many parents have realized that overprotection can affect the growth of their children, especially their only child. At the same time, it is important to realize that being too strict can also negatively affect child development. In educational practice, a principled and emotionally warm parenting approach can help reduce problem behaviors in children.

The results of this study show that experiential avoidance partially mediates the relationship between parenting style and negative emotions, supporting Hypothesis 2. When college students cannot accept and understand the parenting style, they choose to avoid the unpleasant experience as much as possible. High levels of experiential avoidance are a major trigger for negative emotions such as depression and anxiety (Dickson & MacLeod, 2004; Hayes et al., 2004), making it not a solution to the problem. The corresponding positive psychological strategy is cheerful acceptance of parenting styles. As the level of experiential avoidance decreases, the individual's positive emotional experience is enhanced. In family education, parents should avoid using rejection and forced compliance to inhibit their children's emotional expression, as these behaviors can affect their children's level of experiential avoidance and their mental health. Therefore, in school education, negative emotional experiences such as anxiety, depression, and stress caused by an individual's family parenting style can be alleviated by enhancing college students' psychological flexibility, especially their ability for acceptance. In clinical interventions, negative emotions caused by inappropriate parenting styles are difficult to intervene as it relies on the parents. As a result, interventions should focus on seeking more operationally meaningful mediating factors, such as interventions targeting experiential avoidance by college students.

This study demonstrated the effect of negative parenting styles of fathers and mothers on experiential avoidance and negative emotions, verifying Hypothesis 3. In family education, a mother's negative parenting style can influence a father's negative parenting style, in turn affecting experiential avoidance and negative emotions of college students. Family systems theory suggests that a family is an emotional unit. The activities of family members interact with each other, and there is an interaction between the parenting styles of mothers and fathers. This theory implies that, as the main driving force in family education, the mother's parenting style has more influence on family members. Therefore, a negative parenting style should be avoided as much as possible.

In addition, under different family and cultural backgrounds, parents have different ways of raising their children, which can be attributed to social customs, family traditions, educational concepts, and other factors. Past studies showed that parenting style varies in accordance with social and cultural reference groups (Dornbusch et al., 2016; Parra et al., 2019). These studies suggest that society and culture play an important role in the influence of parenting styles on children, mainly because parenting styles, attitudes, and behaviors are constructed in different historical and cultural contexts (Harkness & Super, 1992; Pinquart & Kauser, 2018). In Chinese culture, mothers usually take more responsibility for housework and taking care of children, and mothers often play an important role in children's education. They may pay more attention to children's daily life care, emotional care, and psychological development, and tend to give children more warm, considerate, and meticulous education. In this case, the father's parenting style is often influenced by the mother's, and the father may adjust his own attitudes and behaviors according to the mother's parenting style. Hence, rather than attempting to provide a multicultural perspective on parenting styles, the current study focused on influences within the family between mothers and fathers. However, this difference is not universally applicable to all Asian countries, as the Asian region has diverse cultures and values.

In this study, we tested the hypothesis regarding the chain medication effect of paternal parenting style and experiential avoidance, in relation to mother's parenting style on negative emotions, expanding on past studies on parenting style. However, there are some limitations to the current study. First, the measurement tools need to be further improved. This study adopts the recall method to measure the parenting style of college students, and the absence of instructions to participants about which period of parenting they should recall is a limitation. This may be due to the lack of explicit inquiry in the original scale regarding the specific period of growth experience recalled by participants. The limitations of recalling parenting styles have been discussed in previous studies (Rothrauff et al., 2009). One study noted that retrospective measurements "may constitute valid indicators of the individual's current perception of those features, and as such, may be useful in understanding psychological development or adjustment" (Henry et al., 1994). While this may be a challenge that cannot be resolved at the present stage, future studies will strive to employ diversified measures by integrating subjective reports from both parents and children when assessing parenting style. Secondly, the representativeness of the sample needs to be further improved. Future studies could

further expand the sample size, use more scientific methods to calculate the sample size (Lakens, 2022; Schoemann et al., 2017), and adopt stratified sampling for participant recruitment. Given that participants in this study were from the same university, the applicability of the results of this survey needs to be further verified. Readers need to be cautious if they plan to generalize the results of the survey outside the survey institution. In addition, there are limitations in conducting cross-sectional studies with college students as participants. Future studies can extend the range of participants, conduct longitudinal follow-up surveys, and examine the interactive effects of an individual's psychological traits on parenting styles.

Conclusion

In conclusion, this study explains the psychological mechanism of parenting styles of mothers and fathers influencing college students' emotional health in the context of Chinese culture. We demonstrated that parenting styles have a direct effect on the negative emotions of college students. Experiential avoidance may be the most relevant mechanism accounting for these associations, as an intermediary between two variables. At the same time, the negative parenting style of mothers will affect the negative parenting style of fathers, explaining the internal influence mechanism of parenting styles within the family.

AUTHOR'S CONTRIBUTION

J.W. conducted the analyses and wrote the first manuscript; J.W. and S.F. conceived the current work, revised the manuscript and confirmed its final version.

CONFLICT OF INTEREST

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest. The authors declare no conflict of interest.

REFERENCES

Alonso-Stuyck, P. (2019). Which parenting style encourages healthy lifestyles in teenage children? Proposal for a model of integrative parenting styles. *International Journal of Environmental Research and Public Health*, *16*(11), Article 2057. https://doi.org/10.3390/ijerph16112057

Argyriou, E., Bakoyannis, G., & Tantaros, S. (2016). Parenting styles and trait emotional intelligence in adolescence. *Scandinavian Journal of Psychology*, *57*(1), 42–49. https://doi.org/10.1111/sjop.12266

Asici, E., & Sari, H. I. (2021). A proposed model to explain happiness in college students: The roles of perceived parenting styles, emotional self-efficacy, and forgiveness. *Journal of Adult Development*,

28(4), 332–345. https://doi.org/10.1007/s10804-021-09378-0

Auty, S. G., Lipson, S. K., Stein, M. D., & Reif, S. (2022). Mental health service use in a national sample of college students with co-occurring depression or anxiety and substance use. *Drug and Alcohol Dependence Reports*, *2*, Article 100025. https://doi.org/10.1016/j.dadr.2022.100025

Baardstu, S., Karevold, E. B., & von Soest, T. (2017). Childhood antecedents of Agreeableness: A longitudinal study from preschool to late adolescence. *Journal of Research in Personality*, *67*, 202–214. https://doi.org/10.1016/j.jrp.2016.10.007

Bond, F. W., Hayes, S. C., Baer, R. A., Carpenter, K. M., Guenole, N., Orcutt, H. K., Waltz, T., & Zettle, R. D. (2011). Preliminary psychometric properties of the Acceptance and Action Questionnaire-II: A revised measure of psychological inflexibility and experiential avoidance. *Behavior Therapy*, *42*(4), 676–688. https://doi.org/10.1016/j.beth.2011.03.007

Bowen, M. (1993). *Family therapy in clinical practice*. Jason Aronson.

Cai, X. B., Gui, S. C., Tang, Y. Q., Zhang, S., & Xu, M. X. (2022). The impact of parenting styles on resilience of adolescent: The chain mediating effect of self-control and regulatory focus. *Psychological Development and Education*, *4*, 505–512. (in Chinese)

Cao, J., Ji, Y., & Zhu, Z. H. (2013). Reliability and validity of the Chinese version of the Acceptance and Action Questionnaire-Second Edition (AAQ-II) in college students. *Chinese Mental Health Journal*, *11*, 873–877. https://doi.org/10.3969/j.issn.1000-6729.2013.11.014 (in Chinese)

Children's Work Department of Chinese National Woman Association Union. (2015, December 23). The results of the second national survey on the status of family education were released. CCTV.com. http://news.cntv.cn/2015/12/23/ARTI1450841924965448.shtml (in Chinese)

Coplan, R. J., Hastings, P. D., Lagace-Seguin, D. G., & Moulton, C. E. (2002). Authoritative and authoritarian mothers' parenting goals, attributions, and emotions across different childrearing contexts. *Parenting: Science and Practice*, *2*(1), 1–26. https://doi.org/10.1207/S15327922PAR0201_1

Coplan, R. J., Reichel, M., & Rowan, K. (2009). Exploring the associations between maternal personality, child temperament, and parenting: A focus on emotions. *Personality and Individual Differences*, *46*(2), 241–246. https://doi.org/10.1016/j.paid.2008.10.011

Darling, N., & Steinberg, L. (1993). Parenting style as context: An integrative model. *Psychological Bulletin*, *113*(3), 487–496. https://doi.org/10.1037/0033-2909.113.3.487

Dickson, J. M., & MacLeod, A. K. (2004). Approach and avoidance goals and plans: Their relationship to anxiety and depression. *Cognitive Therapy and Research*, *28*(3), 415–432. https://doi.org/10.1023/B:COTR.0000031809.20488.ee

Dornbusch, S. M., Ritter, P. L., Leiderman, P. H., Roberts, D. F., & Fraleigh, M. J. (2016). The relation of parenting style to adolescent school performance. In R. M. Lerner & J. Jovanovic (Eds.), *Cognitive and moral development, academic achievement in adolescence* (pp. 276–289). Routledge.

Fang, P., Xiong, D., & Guo, C. (2003). The effect of parenting styles on children's academic achievement. *Journal of Psychological Science*, *1*, 73–76. https://doi.org/10.16719/j.cnki.1671-6981.2003.01.021

Gong, X., Xie, X.-Y., Xu, R., & Luo, Y.-J. (2010). Psychometric properties of the Chinese versions of DASS-21 in Chinese college students. *Chinese Journal of Clinical Psychology*, *4*, 443–446. https://doi.org/10.16128/j.cnki.1005-3611.2010.04.020

Gugliandolo, M. C., Mavroveli, S., Costa, S., Cuzzocrea, F., & Larcan, R. (2019). The relative contribution of parenting practices in predicting trait emotional intelligence in an Italian adolescent sample. *British Journal of Developmental Psychology*, *37*(4), 585–599. https://doi.org/10.1111/bjdp.12299

Harkness, S., & Super, C. M. (1992). Parental ethnotheories in action. In I. E. Sigel, A. V. McGillicuddy-DeLisi, & J. J. Goodnow (Eds.), *Parental belief systems: The psychological consequences for children* (2nd ed., pp. 373–391). Lawrence Erlbaum.

Hayes, A. F. (2013). *Introduction to mediation, moderation, and conditional process analysis: A regression-based approach*. Guilford Press.

Hayes, S. C. (2004). Acceptance and commitment therapy, relational frame theory, and the third wave of behavioral and cognitive therapies. *Behavior Therapy*, *35*(4), 639–665. https://doi.org/10.1016/S0005-7894(04)80013-3

Hayes, S. C. (2016). Acceptance and Commitment Therapy, Relational Frame Theory, and the third wave of behavioral and cognitive therapies – Republished article. *Behavior Therapy*, *47*(6), 869–885. https://doi.org/10.1016/j.beth.2016.11.006

Hayes, S. C., Luoma, J. B., Bond, F. W., Masuda, A., & Lillis, J. (2006). Acceptance and Commitment Therapy: Model, processes and outcomes. *Behaviour Research and Therapy*, *44*(1), 1–25. https://doi.org/10.1016/j.brat.2005.06.006

Hayes, S. C., Strosahl, K., Wilson, K. G., Bissett, R. T., Pistorello, J., Toarmino, D., Polusny, M. A., Dykstra, T. A., Batten, S. V., Bergan, J., Stewart, S. H., Zvolensky, M. J., Eifert, G. H., Bond, F. W., Forsyth, J. P., Karekla, M., & McCurry, S. M. (2004). Measuring experiential avoidance: A preliminary test of a working model. *The Psychological Record*, *54*(4), 553–578. https://doi.org/10.1007/BF03395492

Henry, B., Moffitt, T. E., Caspi, A., Langley, J., & Silva, P. A. (1994). On the "remembrance of things past": A longitudinal evaluation of the retrospective method. *Psychological Assessment*, *6*(2), 92–101. https://doi.org/10.1037/1040-3590.6.2.92

Jiang, J., Lu, Z.-R., Jiang, B.-J., & Xu, Y. (2010). Revision of the short-form Egna Minnen av Barndoms Uppfostran for Chinese. *Psychological Development and Education*, *26*(01), 94–99. https://doi.org/10.16187/j.cnki.issn1001-4918.2010.01.017

Kılıçkaya, S., Uçar, N., & Denizci Nazlıgül, M. (2023). A systematic review of the association between parenting styles and narcissism in young adults: From Baumrind's perspective. *Psychological Reports*, *126*(2), 620–640. https://doi.org/10.1177/00332941211041010

Lakens, D. (2022). Sample size justification. *Collabra: Psychology*, *8*(1), Article 33267. https://doi.org/10.1525/collabra.33267

Lerner, R. M., & Busch-Rossnagel, N. A. (Eds.). (1981). *Individuals as producers of their development: A life-span perspective*. Academic Press. https://doi.org/10.1016/C2013-0-11075-6

Lerner, R. M., Lerner, J. V., & Buckingham, M. H. (2023). The development of the developmental science of adolescence: Then, now, next—and necessary. In L. J. Crockett, G. Carlo, & J. E. Schulenberg (Eds.), *APA handbook of adolescent and young adult development* (pp. 723–741). American Psychological Association. https://doi.org/10.1037/0000298-044

Li, J.-B., Willems, Y. E., Stok, F. M., Deković, M., Bartels, M., & Finkenauer, C. (2019). Parenting and self-control across early to late adolescence: A three-level meta-analysis. *Perspectives on Psychological Science*, *14*(6), 967–1005. https://doi.org/10.1177/1745691619863046

Li, Y. (2018). The impact of parental rearing style on learning engagement among senior high school students: A serial mediation effect model. *Psychological Development and Education*, *34*(5), 576–585. https://doi.org/10.16187/j.cnki.issn1001-4918.2018.05.08

Li, Z., Tang, J., & Zhao, N. (2003). A research into the mental health of undergraduates in the transforming period and their parental ways and means in bringing-up. *Journal of Chongqing Normal University (Natural Science Edition)*, *4*, 83–87. (in Chinese)

Lovibond, S. H., & Lovibond, P. F. (1996). *Manual for the depression anxiety stress scales*. Psychology Foundation of Australia.

McKinney, C., Milone, M. C., & Renk, K. (2011). Parenting and late adolescent emotional adjustment: Mediating effects of discipline and gender. *Child Psychiatry & Human Development*, *42*(4), 463–481. https://doi.org/10.1007/s10578-011-0229-2

Monestès, J. L., Karekla, M., Jacobs, N., Michaelides, M. P., Hooper, N., Kleen, M., Ruiz, F. J., Miselli, G., Presti, G., Luciano, C., Villatte, M., Bond, F. W., Kishita, N., & Hayes, S. C. (2018). Experiential avoidance as a common psychological process in European cultures. *European Journal of Psychological Assessment*, *34*(4), 247–257. https://doi.org/10.1027/1015-5759/a000327

Niditch, L. A., & Varela, R. E. (2012). Perceptions of parenting, emotional self-efficacy, and anxiety in youth: Test of a mediational model. *Child & Youth Care Forum*, *41*(1), 21–35. https://doi.org/10.1007/s10566-011-9150-x

Paquette, D. (2004). Theorizing the father-child relationship: Mechanisms and developmental outcomes. *Human Development*, *47*(4), 193–219. https://doi.org/10.1159/000078723

Parra, Á., Sánchez-Queija, I., García-Mendoza, M. D. C., Coimbra, S., Egídio Oliveira, J., & Díez, M. (2019). Perceived parenting styles and adjustment during emerging adulthood: A cross-national perspective. *International Journal of Environmental Research and Public Health*, *16*(15), Article 2757. https://doi.org/10.3390/ijerph16152757

Perris, C., Jacobsson, L., Linndström, H., von Knorring, L., & Perris, H. (1980). Development of a new inventory for assessing memories of parental rearing behaviour. *Acta Psychiatrica Scandinavica*, *61*(4), 265–274. https://doi.org/10.1111/j.1600-0447.1980.tb00581.x

Pinquart, M., & Kauser, R. (2018). Do the associations of parenting styles with behavior problems and academic achievement vary by culture? Results from a meta-analysis. *Cultural Diversity and Ethnic Minority Psychology*, *24*(1), 75–100. https://doi.org/10.1037/cdp0000149

Podsakoff, P. M., & Organ, D. W. (1986). Self-reports in organizational research: Problems and prospects. *Journal of Management*, *12*(4), 531–544. https://doi.org/10.1177/014920638601200408

Rothrauff, T. C., Cooney, T. M., & An, J. S. (2009). Remembered parenting styles and adjustment in middle and late adulthood. *Journals of Gerontology: Series B*, *64B*(1), 137–146. https://doi.org/10.1093/geronb/gbn008

Sari, T., & Ozkan, I. (2016). An investigation of the relationship between adolescents' subjective well-being and perceived parental attitudes. *Dusunen Adam Journal of Psychiatry and Neurological Sciences*, *29*(2), 155–162. https://doi.org/10.5350/DAJPN2016290207

Schoemann, A. M., Boulton, A. J., & Short, S. D. (2017). Determining power and sample size for simple and complex mediation models. *Social Psychological and Personality Science*, *8*(4), 379–386. https://doi.org/10.1177/1948550617715068

Shan, C. H., & Hawkins, R. (2014). Childcare and parenting practices in Singapore: A comparison of fathers' and mothers' involvement. *Journal of Tropical Psychology*, *4*, Article e10. https://doi.org/10.1017/jtp.2014.10

Shrout, P. E., & Bolger, N. (2002). Mediation in experimental and nonexperimental studies: New procedures and recommendations. *Psychological Methods*, *7*(4), 422–445. https://doi.org/10.1037/1082-989X.7.4.422

Starr, L. R., Shaw, Z. A., Li, Y. I., Santee, A. C., & Hershenberg, R. (2020). Negative emotion differentiation through a developmental lens: Associations with parental factors and age in adolescence. *Personality and Individual Differences*, *152*, Article 109597. https://doi.org/10.1016/j.paid.2019.109597

Steinberg, L. (2001). We know some things: Parent-adolescent relationships in retrospect and prospect. *Journal of Research on Adolescence*, *11*(1), 1–19. https://doi.org/10.1111/1532-7795.00001

Tabor, E., Patalay, P., & Bann, D. (2021). Mental health in higher education students and non-students: Evidence from a nationally representative panel study. *Social Psychiatry and Psychiatric Epidemiology*, *56*(5), 879–882. https://doi.org/10.1007/s00127-021-02032-w

Wang, L., & Fu, J. Z. (2005). A summary of parenting style and children's development in China. *Advances in Psychological Science*, *13*(03), 298–304.

Wen, H., Ma, H., & Zhang, C. (2021). The relationship between college students' parental rearing style and mental health: The mediating role of security. *Chinese Journal of Health Statistics*, *2*, 226–228. (in Chinese)

Wolfradt, U., Hempel, S., & Miles, J. N. V. (2003). Perceived parenting styles, depersonalisation, anxiety and coping behaviour in adolescents. *Personality and Individual Differences*, *34*(3), 521–532. https://doi.org/10.1016/S0191-8869(02)00092-2

Zhang, H., Tian, L. C., Wu, B. S., & Wang, Z. H. (2021). The influence of parental rearing style on interpersonal adjustment of college freshmen: The mediating role of psychological capital. *Survey of Education*, *10*(13), 9–12+25. https://doi.org/10.16070/j.cnki.cn45-1388/g4s.2021.13.003 (in Chinese)

Zhang, J. R., Meng, F. F., Ling, H., Gong, W. T., & Li, J. X. (2021). Relationship among childhood abuse, parenting styles, attachment, and antisocial personality disorder in college students. *Chinese Journal of Clinical Psychology*, *29*(01), 28–32. https://doi.org/10.16128/j.cnki.1005-3611.2021.01.006 (in Chinese)

Zhang, W.-X., & Chen, G.-H. (2009). Developmental Contextualism: An instance of development system theories. *Advances in Psychological Science*, *17*(04), 736–744.

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